



Analysing role of businesses' investment in digital literacy: A case of Pakistan

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ABSTRACT

The digital economy in Pakistan holds abundant potential, yet its existence is far from commonplace for ordinary Pakistani citizens. Understanding and implementing strategies to expedite the onboarding of masses onto digital platforms is paramount for Pakistan's digital economy. This study aims to examine and confirm whether digital transactions can be increased by improved digital literacy of customers. The idea is to find whether companies that invest in digital literacy can reap benefits in the form of increased transactions. The novelty of this research topic lies in the fact that research about digital literacy and the digital economy, especially in association with one another, is scarce worldwide. The subject provides a theoretical foundation for developing countries like Pakistan, where digital technologies, unsatisfactory literacy rates, and low internet penetration exist. The study uses multiple linear regression models and found that digital literacy and digital media usage enablers influence digital sales and transactions in Pakistan. The practical implication of the finding can be linked with digital economy development in Pakistan in the form of the establishment of various new apps and businesses in transportation (Careem, Bykea), food (Foodpanda) and social media (TikTok).

1. Introduction

1.1. Background

Increasing growth in information and communication technologies paved the way for reaching potential customers or people who may eventually become customers (Acosta-Prado and Tafur-Mendoza, 2021). Firms do strategize to improve their performance by making investments to increase their customer base because firms mainly earn through their customer base and sales. Therefore, firms adopt several strategies to get a larger number of customers, but selecting the right strategy is based on many factors. For instance, a firm may advertise via communication media or through in-person promotions, amongst other options. Such strategies aim to aware customers (or potential customers) about firms products or services.

Most firms make a significant investment in promoting their products (Van Triest, et al., 2009). Such investment become crucial when the product (or service) is relatively new in the market. Sometimes, the customers may be aware of the product but may lack detailed insights into utilizing it. This is significant for products for which tech-savvy is a

prerequisite. For instance, Microsoft Excel is a fundamental tool for data scientists, but someone who cannot use computers may find MS Excel a rather expert-level tool.

The digital economy and its associated business also fall in the category where customers need technical knowledge (e.g., Brynjolfsson & Kahin (2002) who call the digital economy a driver of economic and social change through information, computing, and communications). The term digital economy is a broad term that engulfs all businesses that operate over digital media or are facilitated. With fast-paced developments in the technology sector, the entire world is shifting towards a digital platform with agility (Dahlman et al., 2016). Digital technology has revolutionized the way business is done, uprooting legacy systems and replacing them with quicker, more efficient, cost-beneficial digital systems. With the onset of COVID-19, the significance of isolation has further alleviated the utilization of digital systems. There was about a 12% growth in active social media users by October 2020 versus the last 12 months (Kemp, 2021). Also, "... on average of more than 14 people started using social media every second..." (Kemp, 2021). The advent of the virus led to masses flocking towards the digital world for comfort, convenience, and necessity. With the increase in digital users, the digital

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economy has seen a boom in potential customers. Currently, digital business owners have a vast sea of opportunities to capitalize. And even though a much larger number of people are presently digitally aware, there is still a considerable part of the world's population that is severely unaware of digital technologies.

1.2. Digital literacy in Pakistan

Pakistan is a country with over 220 million inhabitants (PBS, 2018). The literary rate has been consistently rising, but the national literacy rate is still around 60% (MoF, 2021). The year 2020 has seen a blatant rise in mobile phone use for the Pakistani population, but there is still a long way to go (Zahra-Malik, 2020). The mobile phone connection subscriptions have been on steady growth too in the last two decades. Fig. 1 depicted the growth rate of mobile phone subscriptions in Pakistan during 2005–2019. Also, January 2021 marked close to 60 million internet users in Pakistan (Kemp, 2021). The growth in internet usage was 21% between 2020 and 2021 and 17% between 2020 and 2019. With an estimated internet penetration rate of 27%, Pakistan has a long way to go; the journey may be long and arduous, but it seemingly holds proliferous economic potential.

Pakistan's digital economy is steadily growing, especially with increased government-funded incubators that help launch digital start-ups. The digital scene here has seen exponential growth in the last decade in particular. However, Pakistan's is also facing a lack of awareness about digital business due to low literacy rate and internet usage). Although digital content in the educational curriculum may help, its long-term investment may not be possible to adopt for populations out of school. Therefore, currently, Pakistan is home to many non-tech-savvy residents who offer a large customer base for the digital business. However, digital firms fail to capitalize on this potential due to a lack of awareness of their potential customers about digital business and the products (services it offers). One way of capturing these potential customers is to engage them in innovative ways for increasing their awareness about digital business. In this regard, some local digital firms such as Foodpanda, Careem, Daraz, and Bykea can engage their potential customers and increase their customer base by raising awareness about their business.

This study explores Pakistani digital firms working towards digital literacy. The study focuses on two objectives. First, its study impacts of digital firm's investment in different skills of its customers on their sales. Secondly, it aims to identify enablers for digital media usage. The study uses primary data collected from over 50 firms working in Pakistan for the former objective, whereas it uses secondary data from government official sources for the later goal. We constructed several hypotheses based on our theoretical framework. The study uses multiple linear regression models. The study contributes to the scholarly literature on

the digital economy and literacy in two ways. First, it is the first study of its kind on the digital economy of Pakistan (as earlier studies like e.g., Mujahid, 2002; Zahid et al., 2012; Malik and Wahaj, 2019, does not touch this aspect of the digital economy). Second, the study provides new insights to understand how digital firms can increase their customers base further (especially in developing countries) by adopting strategies related to investment in the awareness of its potential customers through different digital skills as earlier studies like e.g., Bell & Eisingerich (2007) and Bell et al. (2017) are more focused on a single aspect of customers digital learning.

The rest of the study is organized as follows. The following section is an overview of associated literature and establishing a base for underlying hypotheses. Section 3 specifies the methodology applied to the collected data and how econometric models are developed. Section 4 presents results and a discussion. The final section concluded the study.

2. Literature review

2.1. Digital economy

Over the years, many researchers have explored what is meant by the term *digital economy*. Bountiful publications exist that address what it means to be operating in the digital economy in their respective eras. The accelerated growth of technology can be thanked for the ever-changing definitions of the digital economy. The last two decades, in particular, have been the hallmark of development in this sector. The advent of the internet and everything else that followed has continually added to the definition of the *digital economy*.

Scholarly literature presents a wide variety of definitions for the digital economy; most definitions describe it differently based on its operation, but the general theme remains the same. For instance, Mesenbourg (2001) defined the digital economy as e-commerce, e-businesses, and all the digital technologies available to facilitate the aforementioned business methods. Whereas, Carlsson (2004) emphasized that this 'new age' economy is all about dynamic efficiencies instead of static ones. Furthermore, Brynjolfsson & Kahin (2002) describe the *digital economy* as a driver of economic and social change via information, computing, and communications. Similarly, Haltiwanger & Jarmin (2002) call the *digital economy* the "information economy," emphasizing the significance of incorporating the right kind of information in driving business decisions. Borremans et al. (2018) explore the concept of business strategy in the digital economy and suggests that an IT-based business model uses people, technology, financial instruments, and business strategy altogether. To sum it up, all economic activities that employ Information and Communication Technology (ICT) must fall under the digital economy (Bukht and Heeks, 2017).

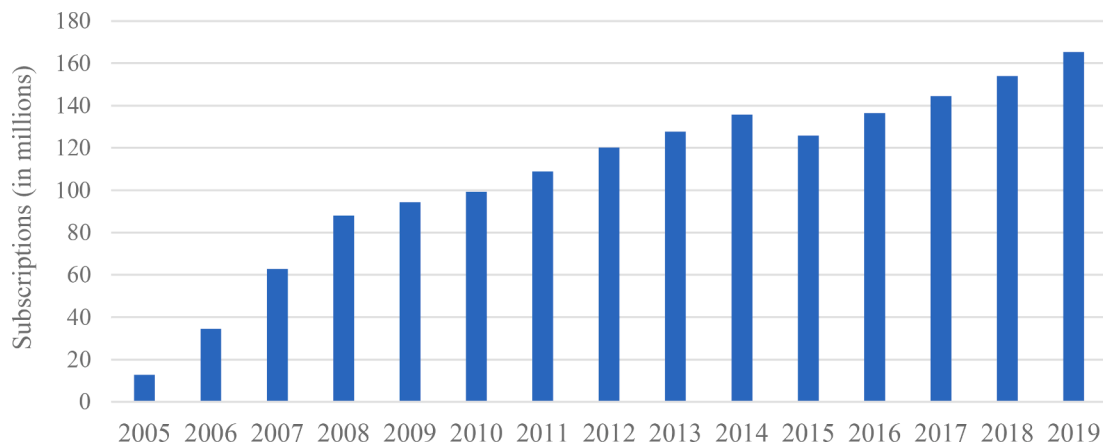


Fig. 1. Pakistan's Mobile-phone subscriptions growth over the years (ITU, 2020).

Interestingly, despite the abundance of scholarly work on the *digital economy*, studies about Pakistan have been scarce. Some limited available studies do explore a few aspects of the digital economy in Pakistan. For instance, [Mujahid \(2002\)](#) investigated the “digital divide” and the future of Pakistan’s digital economy theoretically. The study examined various economic sectors that could benefit from ICT use and can benefit Pakistan’s economy. [Zahid et al. \(2012\)](#) explored the relationship between customer satisfaction and econometric banking in Pakistan. More recently, [Malik & Wahaj \(2019\)](#) examined the social impact of the digital economy, particularly in ride-sharing platforms.

Although the studies above are exceedingly essential topics that need to be addressed, however, they do not discuss the current situation of the entire *digital economy*, i.e., ignoring the macroeconomic aspect of the *digital economy*. Further, these studies do not address the essence and the root of Pakistan’s most significant obstacle in the *digital economy* establishment, i.e., establishing a commonplace knowledge for digital literacy. The establishment of a strengthened *digital economy* beseeches for a mass understanding of digital media and how to use them. Herein lies the research gap which we aim to fulfil via this study.

2.2. Digital literacy

The other central concept addressed by this study is that of digital literacy. Digital literacy refers to the digital literacy of the masses who are end-users of digital economy businesses or their potential customers. In contrast to *digital economy* definitions, the concept of digital literacy is less defined and less standardized. The definition of digital literacy varies in most of the scholarly work based on the concept or idea the researchers intended to explore in that particular research. For example, [Glistler \(1997\)](#) defined digital literacy as human interaction with the World Wide Web. More recent studies included more advanced technological components in determining digital literacy ([Buckingham, 2010](#); [Bawden, 2008](#)). However, a key feature in all existing definitions is that digital literacy is defined in the context of human experiences ([Alkali and Amichai-Hamburger, 2004](#); [Martin, 2008](#)).

More recently, the rise of the COVID-19 pandemic and subsequent shift to online teaching and the extensive use of video conferencing applications since 2020 are examples of growing digital literacy in academia and work. Before the COVID pandemic, education via digital platforms existed but not in a mainstream manner ([Sugiantari, 2020](#)). The pandemic and the subsequent social isolation led to the transition of teaching to digital media. Although associated personnel and students struggled for a while, eventually, there was a vast surge in the usage of video conferencing tools for educational purposes ([Daniel, 2020](#)). It is especially true for developed countries where relevant technologies are accessible by most. The sudden increase in employment tools such as Zoom, Google Classrooms, and Discord led the masses to adapt and learn to use these tools ([Allen et al., 2020](#)). It is one case where necessity and need drove the adaption of digital literacy skills; however, the success of this model depends on the availability of digital technologies to the masses (especially in countries like Pakistan, where not all areas have internet access). Thus, the model may only work in limited scenarios and does not consider outside of classroom learning.

Different scholars pointed towards various similar components via which digital literacy is defined. For example, [Dobson & Willinsky \(2009\)](#) described digital literacy through word processing, writing, linking text and information, and widespread access to information. Despite the variety of definitions of digital literacy, most of them attempt to avoid overlap. For the sake of simplicity, we have selected the factors of digital literacy as defined by the United Nations (UN) in their Sustainable Development Goals (SDGs) for digital literacy ([Law et al., 2018](#)). We segregated some of UN’s digital literacy components based on the overlap. It was done on the recommendation of business studies and entrepreneurship experts who claimed that a customer-centric approach would require a slightly varying view instead of the UN one. They explained that the SDG was not a business plan but more of a

development program. Therefore, the digital literacy in digital businesses would be minutely different. The list of digital literacy components thus obtained is listed as functional skills (DL₁), creativity skills (DL₂), information literacy (DL₃), critical thinking and evaluation (DL₄), communication skills (DL₅), collaboration skills (DL₆), e-safety (DL₇), and problem-solving skills via digital technologies (DL₈).

One of the objectives of this study is to explore whether digital firms should invest in digital media awareness and education of their target audience or customer base. To analyse the link between the *digital economy* and digital literacy, the above eight digital literacy competencies (DL₁ – DL₈) are tested as independent variables against the number of transactions in the digital economy as the dependant variable. These eight competencies with the constructed hypotheses are depicted in the following sub-sections of the literature review. For the formulation of the research framework, a blanket hypothesis is drafted that associates digital literacy as a whole against growth in transactions, where the latter is the independent variable while the former is dependant. The hypothesis is given as follows:

H_A: *Investing in the digital literacy of customers has a positive influence on the number of transactions.*

This main hypothesis lead to the development of more hypotheses using contemporary scholarly literature on the subject matter as is described in detail in the following sub-section.

2.3. Investment in digital literacy of customers

The *functional skills* refer to the basic operation of a digital device. Unimaginable in 2018, Pakistani communities that had limited access to digital media, are now active smartphone users. According to the HootSuite report on Pakistan’s digital ecosystem, the yearly growth in mobile connections by January 2021 was 6.9 million, i.e., about a 4% increase in users ([Kemp, 2021](#)). At the same time, social media users skyrocketed by 9 million, a 24% increase ([Kemp, 2021](#)). These facts are indicators that the Pakistani audience is slowly moving towards a more digital paradigm. Thus, the *hypothesis (H₁)* stands to be whether companies who tried to improve their target audience’s functional skills successfully increased their outputs.

Creativity as a digital literacy component can create fresh content and make use of digital technologies ([Nottingham Trent University, 2021](#)). According to [Hartley et al. \(2008\)](#), the mainstream use of digital technologies, creative content creation was thought to be profession-specific and requiring expertise. This study emphasizes that creativity has long been a driver in economic growth. The onset of technological innovation has given way to a more customer-centric and user-led approach to content creation. It can be related to Pakistan’s digital landscape, where a previously television-driven entertainment model has been heavily disrupted by digital content creation ([Malik, 2020](#)). Therefore, the hypothesis (H₂) explores that investing in the target audience’s creativity benefits the firms.

Information sifting (or retrieval) and selection for decision-making are often termed information literacy ([Koltay, 2011](#)). The ability and skill to receive information from a digital platform, perceive it correctly, and base decisions on this premise are the idea behind information literacy. Subsequently, the following hypothesis (H₃) seeks to find whether companies that invest in the information literacy of their audience can capitalize on it.

According to [Glistler \(1997\)](#) and [Lankshear & Knobel \(2008\)](#), the idea of digital literacy is not limited to just the use of digital technologies. They suggest that digital literacy also critically assesses ideas and items on digital platforms and evaluates them as legit or valuable. Both studies condone the idea that digital literacy comprises of critical evaluation of information present on digital media. Furthermore, the concept is limited to the critical assessment of information alone and includes all available ideals and items. It may even be considered a vastly complex skill that is difficult to acquire. Based on this discussion, we formulate a *hypothesis (H₄)* that firms who invest in establishing *critical thinking* and

evaluation skills of customers reap its benefits.

Communication research is bountiful, and thus digital literacy of communication skills has mainly been explored over the years. Therefore, more recent studies focus more on digital communication's propensity and the risks associated with it (Nasah et al., 2010). Other researchers have explored how to move a step further from communication into the world of training and knowledge sharing (Gómez-Trigueros et al., 2019). However, it is notable that the established studies carried out in communication are more on the communication aspect than on digital literacy and digital economy benefits. It leads to *hypothesis* (H_5) that seeks to confirm if firms have benefitted from investing in the establishment and strengthening of *communication skills* of their audience.

According to Hobbs & Coiro (2016), activities of teamwork and collaboration over digital media lead to a better grip on digital literacy. Genlott & Grönlund (2016) indicated that *collaboration* is a skill that can help younger audiences as well. Although several scholarly studies do address that how collaborative experience and digital technology can lead to improved competence in digital literacy as a whole (Sharp, 2018), most of these are instructor-focused (specific to educationalists can benefit through the use of collaborative digital experiences) (Mills, 2018). For example, Techataweewan & Prasertsin (2018) reported collaboration as an important segment of digital literacy deemed necessary for Thai students. The lack of focus in scholarly work towards the commercial viewpoint of digital literacy improvement strategies, particularly of collaborative digital literacy, fuels the establishment of the *hypothesis* (H_6) that aims to measure whether investments in collaboration skills impact a firm's sales.

Electronic media's data security and safety are vital components of technological literacy (Burkell et al., 2015). The significance of *e-safety* is especially emphasized in the dialogue on children's digital literacy (Hague and Payton, 2011) or higher education students (Nelson et al., 2011). There is extensive literature on cyber-security with associated risks (e.g., Cirmu et al., 2018) or on improving cyber-security and *e-safety* (e.g., Harrop and Matteson, 2015; Petrenko, 2018; Pupillo, 2018). A recent study by Chang & Coppel (2020) discusses the awareness of the masses about cyber-security in Myanmar. However, the understanding of customers about *e-safety* and security is scarcely addressed. It set a foundation for our *hypothesis* (H_7) to find the link between investments in *e-safety* awareness of customers and firms' sales.

Problem-solving through digital technologies is a relatively advanced digital literacy skill (Feola, 2016). Compared to the more baseline digital literacy skills, problem-solving requires advanced digital tools that facilitate problem-solving (Ozdamar-Keskin et al., 2015). The complex nature of this competency set the foundation for *hypothesis* (H_8) that investigates if digital companies' investment in building the problem-solving skills of their customer can benefit these firms.

2.4. Digital media usage

In addition to the digital literacy components, the study also focuses on parameters that may be considered enablers of digital media usage. The selection of the enablers is based on the available data on Pakistan's various vital statistics. These factors are internet usage percentage (IN), fixed broadband subscriptions (FB), mobile-cellular telephone subscriptions (MB), and literacy rate (LI). It is essential to mention that this is not the exhaustive list of digital media enablers but is limited to factors for which comprehensive data was readily available for Pakistan for ten year period.

This study considers these factors under observation and analyses the available statistics to determine whether they influence sales or transactions in the *digital economy*. Thus a second central hypothesis (HB) is developed as follows:

H_B: *Digital media usage enablers influence the number of digital transactions in Pakistan.*

Based on H_A and H_B, a theoretical framework is established, as presented in Fig. 2. This theoretical framework has two components. The first component is focused on the role of digital literacy by the firms (that could be measured through eight factors) and its influence on digital transactions. This part is covered through primary data (about which details are presented in the subsequent section).

2.5. Enablers of digital media

The second part, which focuses on the influence of digital media enablers, is covered through secondary data (as explained above). Based on these two parts, 12 sub-hypothesis are constructed (Table 1) and subsequently covered through literature in the rest of this section. This research incorporates secondary data on parameters that may be considered enablers of digital media usage. These enablers are described as follows to develop several hypotheses using contemporary scholarly literature.

The first enabler of digital media usage is *internet access*. Although Pakistan joined the race a little late, the country's internet and digital media ecosystems have grown exponentially over a short period. Internet works as a gateway to the wide world of digital transactions, even though it does not work as a limitation but serves as a facilitator for most digital transactions (Tayibnapis et al., 2018). Therefore, we *hypothesize* (H_9) that internet access in the country impacts the digital economy via digital transactions.

Mobile-phone subscriptions and *broadband connections* are two other enablers that act as gateways to the digital world. Both mediums also enable users to make digital transactions and thus create opportunities in the digital economy. Based on the significance of mobile-phone subscriptions and fixed broadband connections, the subsequent two *hypotheses* (H_{10} and H_{11}) seek the impact of mobile-phone subscriptions and broadband connections (respectively) on digital transactions.

Finally, the fourth enabler is the *literacy rate*. Since there are no actual data points in Pakistani educational reporting for digital literacy, we considered the overall literacy rate and how its growth has impacted the transactions over a digital medium. Thus, the last *hypothesis* (H_{12}) aims to test whether literacy rate affects digital transactions.

3. Data sources and research methodology

This section discusses the data, its sources and then the research methodology used in the study. The objective of the study may be clubbed into studying the link between digital firms investment in digital literacy and its impact on their business transactions and also studying the digital literacy enablers and their link with businesses. These two objectives require a different kind of data, therefore, this study uses two kinds of data, as described in the subsequent section.

3.1. Data and its sources

To study the link between digital firms investment in digital literacy and their business transaction, the study needs primary data from firms. The primary data was collected via a questionnaire survey that aims to test the opinions and experiences of digital firms. The questionnaire survey was drafted to gauge whether firms invest in strengthening the digital literacy of their customers. The survey also aims to find out if this strategic decision helps digital firms increase their customer transactions. The survey contains questions related to respondent's designation, the type of industry (e.g., textile, transportation, communication), size of the firm and its target area, firms investment in all eight skills (DL1 – DL8), and if these eight skills have any impact on the sales.

The survey was conducted online (during January–April 2021) using LinkedIn; digital business owners, executives, and decision-makers were contacted to share their experiences in strategic planning of digital literacy for their customers. To ensure the integrity of exclusive responses,

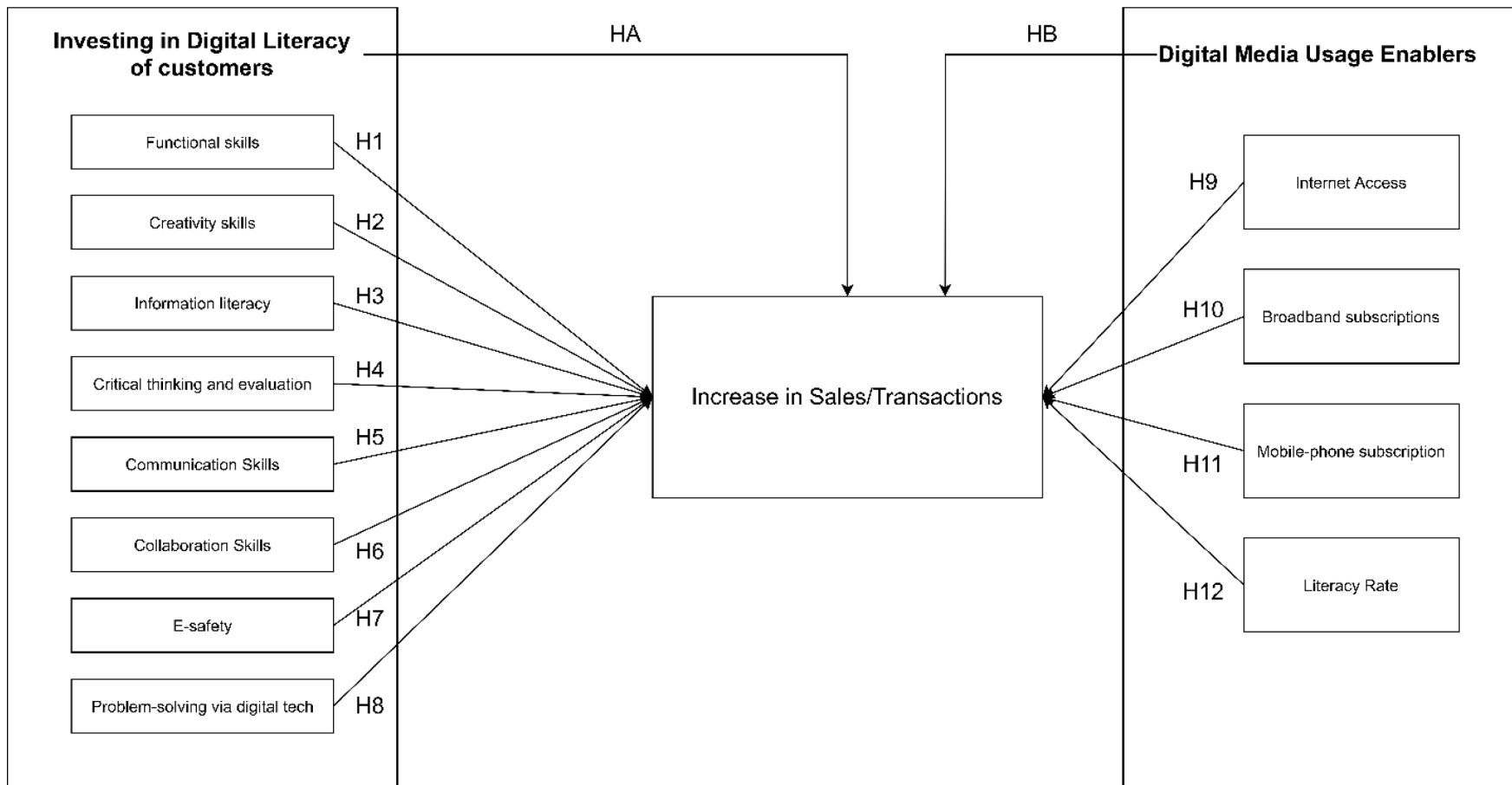


Fig. 2. Theoretical framework.

Table 1
Hypothesis.

Hypothesis
H ₁ : Investing in <i>functional skills</i> of digital literacy of customers positively influences the number of digital transactions.
H ₂ : Investing in the <i>creativity skills</i> of digital literacy of customers influences the number of digital transactions.
H ₃ : Investing in the <i>information literacy</i> of customers has a positive influence on the number of digital transactions.
H ₄ : Investing in <i>critical thinking and evaluation skills</i> of digital literacy of customers has a positive influence on the number of digital transactions.
H ₅ : Investing in <i>communication skills</i> of digital literacy of customers positively influences the number of digital transactions.
H ₆ : Investing in <i>collaboration skills</i> of digital literacy of customers positively influences the number of digital transactions.
H ₇ : Investing in <i>e-safety skills</i> of digital literacy of customers positively influences the number of digital transactions.
H ₈ : Investing in <i>problem-solving skills</i> of digital literacy of customers positively influences the number of digital transactions.
H ₉ : <i>Internet usage</i> influences the number of digital transactions.
H ₁₀ : The number of <i>broadband subscriptions</i> influences the number of digital transactions.
H ₁₁ : The number of <i>mobile phone subscriptions</i> influences the number of digital transactions.
H ₁₂ : <i>Literacy rate</i> influences the number of digital transactions.

a single representative from the firm was contacted. All the respondents were from companies serving the Pakistani market. We reached about sixty Pakistani digital businesses for data collection, out of which 53 responses (88% response rate) were received. The majority of the responses (98%) were from senior management, owner/partner, and executive categories. *Data Reliability* was checked using Cronbach's alpha. It was calculated to be 0.87; since the satisfaction requirement of reliability is 0.7, it can be concluded that the primary dataset is reliable. This data was used to test hypotheses (H₁ – H₈).

To get the second objective of the study, i.e., to study the link between digital literacy enablers and business transactions the study rely on secondary data. The secondary data was retrieved from various sources. These include the Pakistan Bureau of Statistics (data source Pakistan Social and Living Standards Measurement (PLSM)), International Telecommunications Union (ITU Regional/Global Key ICT Indicators), Pakistan Telecommunication Authority (various Annual Reports), and State Bank of Pakistan (report on Payment Systems Review). The data was acquired for the period from 2005 to 2020. However, the information for payment systems is available only for the years 2010–2020. It limits our study to ten year period from 2010 to 2020. The secondary data is used to test hypothesis (H₉ – H₁₂)

3.2. Multiple-linear regression analysis

The methodology applied to both data sets is hypothesis testing using Multiple linear regression models. A generic regression model is established in Eq. (1) as follows:

$$Y = \alpha_1 + \alpha_2 X_1 + \dots + \alpha_n X_n + \varepsilon_i \quad (1)$$

Where Y refers to digital transactions in the economy, α s are the coefficients of interest, ε_i is an error term, and $X_1 - X_n$ consists of two components. The first component of X includes the digital competency levels (eight digital skills shown on the left side of Fig. 2). For measuring the impact of these eight factors on digital transactions, a regression model is given in Eq. (2). The second component includes the enablers for digital literacy (four elements that are shown on the right side of Fig. 2). Accordingly, to study the impact of these digital literacy enablers on digital transactions, a regression model is established as given in Eq. (3).

$$\ln(Y) = \beta_0 + \beta_1 DL_1 + \dots + \beta_8 DL_8 + \mu_i \quad (2)$$

$$\ln(Y) = \gamma_0 + \gamma_1 IN + \gamma_2 FB + \gamma_3 LI + \gamma_4 LR + \mu_i \quad (3)$$

Where DL_1 to DL_8 refers to the digital competencies (functional skills, creativity skills, information literacy, critical thinking and evaluation, communication skills, collaboration skills, e-safety, and problem-solving skills, respectively), IN , FB , MB , and LI refer to the digital enablers (internet usage percentage, fixed broadband subscriptions per 1000 inhabitant, mobile-cellular telephone subscriptions (millions), and literacy rate (percentage), respectively). β s are the associated parameters with each digital competencies, γ s are the related parameters of for digital enablers and μ_i are the standard error terms, respectively.

4. Results and findings

This section is divided into two sub-sections: the first sub-sections present Eq. (2) results and discussion. The section sub-section is based on Eq. (3) results and discussion on it.

4.1. Analysis of digital literacy skills

The regression model of Eq. (2) is estimated using Huber-White robust standard errors consistent procedure in Eviews. The primary data of the analysis was from different firms, and it may have potentially heteroscedasticity issue. However, Breusch-Pagan-Godfrey Test does not show any heteroscedasticity issue (Gujarati, 2011). But, to check the consistency of the results, we used both the Ordinary Least Square method (OLS) and the Huber-White method. The Huber-White process provides heteroskedasticity-consistent standard errors (Gujarati, 2011). The results of both models are presented in Table 2.

The results of both models show a good fit with overall significance demonstrated by statistically significant F-Statistic value and R^2 over 90%. It implies that the models explain over 92% of the total variation in the dependant variable that indicates a good fit. Overall, the model has six statistically significant coefficients with two coefficients (DL_2 and DL_5) statistically not significant. The results imply that investment in customers' digital literacy leads to greater sales of digital businesses (it confirms hypothesis H₄).

The statistically insignificant coefficients of DL_2 and DL_5 imply that firms' investment in customers' creative and communication skills does not enhance digital business sales (rejecting hypotheses H₂ and H₅). The results seem plausible given that investment in creative and communication skills is a long-term investment that customers may not directly acquire such skills through firms' investment for digital literacy. Also, these skills may not necessarily help potential customers to convert into sales or transactions. Additionally, creativity is something that is nurtured and explored based on an individual's various identities.

Table 2
Results of digital literacy skills analysis.

Variables	OLS		Huber-White	
	Coefficient	Std. Error	Coefficient	Std. Error
Functional skills (DL_1)	0.030**	0.013	0.030*	0.018
Creativity skills (DL_2)	0.006	0.011	0.006	0.011
Information literacy (DL_3)	0.066***	0.015	0.066**	0.022
Critical thinking and evaluation (DL_4)	0.054**	0.018	0.054*	0.029
Communication skills (DL_5)	−0.012	0.015	−0.012	0.014
Collaboration skills (DL_6)	0.103***	0.023	0.103**	0.033
E-safety (DL_7)	−0.038*	0.013	−0.038*	0.019
Problem-solving skills (DL_8)	0.043***	0.010	0.043**	0.021
Constant	0.376***	0.045	0.376***	0.046
Model fit information				
F – Statistics	67.297***		67.297***	
R^2	0.924		0.924	
Adjusted- R^2	0.910		0.910	
N	53		53	

Note: *, ** and *** show statistical significance at 10%, 5% and 1% respectively.

Furthermore, most digital firms do not require their clientele to be particularly creative. On the other hand, communication skills are complex phenomena and are closely bonded with language skills. But, language skills are linked with factors, including culture, social status, educational background, and other identities, thus making it unexplainable for our model. These all may explain why these skills are not the digital literacy competencies that digital firms want to invest in.

The hypothesis H_1 is confirmed from the statistically significant value of functional skills, indicating that customers' functional skills of digital literacy do influence digital business sales. These results are also plausible because basic information about digital devices (such as mobile phones, digital pads, or computers) could encourage the customers to use them for shopping or other services being offered by the digital business. It implies that investment in these skills of customers has the potential to convert in sales. Thus, investment in functional skills by a digital company does influence the sales of digital business.

The hypotheses H_3 and H_4 (about information literacy and critical thinking and evaluation skills, respectively) are confirmed from the results. It indicates that the digital business owners and top management think that investment in customers' skills does improve their sales. Similarly, hypothesis H_6 (collaborations skills), H_7 (e-safety skills), and H_8 (problem-solving skills) are also confirmed, indicating that digital business investment in these skills of customers does increase their sales transactions.

4.2. Results applications to Pakistan digital landscape

These all results are in line with new developments in the digital world of Pakistan, such as the establishment of various new apps in transportation, food, or social media. The following two examples evoke digital businesses, the skills of functional digital literacy, information literacy, e-safety, problem-solving and critical thinking.

In 2016, when *Careem* entered the Pakistani market, ride-sharing or any such apps were far from the norm (Ahmed, 2016). This disruptive technology did wonders for the transportation scene in Pakistan, empowering masses while expanding exponentially over a short period (Javaid et al., 2019). It is safe to assume that when Careem penetrated the market, it established the digital literacy skills needed to enable users. Before introducing Careem in Pakistan, the concept of ride-sharing, widespread use of Google Maps and other related technologies was limited and near non-existent. Careem is one of many examples where disruptive digital businesses forced users (and potential users and non-users) to get up to speed with digital technologies.

Similarly, in 2012, *Foodpanda* was launched in Pakistan for food delivery services that served as a hub of restaurants for online ordering (Dawn, 2016). Unlike, Careem, Foodpanda was not a new app (or a pioneering solution) instead of an improved alternative to an already existing solution. Its services grew from 5000 orders a day (Arshad, 2017) to 15,000 orders a day (Attarwala, 2021). Here it seems that the digital divide was more challenging to cross and comparatively slower to overcome. However, in a country like Pakistan, the common theme is that the concept of digital literacy and critical competencies required to engage in economic activities require firms to invest in systems, activities, and strategies that bring more people on board.

On the other hand, collaboration and creativity skills can be explored in the local context using *TikTok* as an example. TikTok has a simpler, user-friendly, and straightforward user interface (UI) than other social media apps (Hinduja, 2021). While emphasizing the simple user interface, United Nations Educational, Scientific and Cultural Organization (UNESCO) stated that people with no or low digital literacy skills or who belong to low-literate backgrounds might be overwhelmed by fancy or complicated UI (Zelezny-Green et al., 2018).

Designing innovative UI that provides the best user experiences may also be considered one of the most efficient and effective methods in familiarizing users with digital technology slowly and seamlessly. TikTok not only provided users with this opportunity but is also more open

to diversity, making it more inclusive than sophisticated high-brow social media platforms such as Twitter or Instagram (Ahmad and Zelenko, 2020). TikTok enabled thousands of low-income household members to become content creators and to share their narratives with the world. The growth of the video-sharing platform is shrouded in controversy that is far from a conversation on e-safety. Despite the many opinions on the application and its users, it stands to reason that TikTok helped its users learn to be more digitally literate by enabling them to become content creators and collaborators.

The *Bykea* application provides a unique perspective of innovation and customer engagement wherein the transportation start-up empowered individuals from lower economic households (Jamil, 2019). The idea behind Bykea stemmed from Pakistan's gridlocked cities; the firm took a fresh take on digital innovation by creating a multilingual UI. Although English is the official language of Pakistan, it is not a language accessible by all; on the contrary, language is one of the foundations for the class divide in the country. Urdu, the national language of Pakistan, is one of the many indigenous languages of the country far more accessible than English. However, the integration of Urdu in modern technology is somewhat stunted. The strategic position taken by Bykea in this scenario is thought to be one of the reasons for Bykea's apparent success in the country (Media Agility, 2021). Integrating a customer-accessible language to facilitate digital literacy awareness is an innovative method that seems to have had inspiring results. This observation further confirms the research conducted in this research. It also implies that firms can monetize off initiatives for the digital literacy growth of the masses.

Finally, e-safety is also critical for digital business transactions, as confirmed from hypothesis H_7 . It is essential to mention that end-users participate in digital transactions only when he is aware of e-safety. His lack of e-safety understanding will detract from performing digital transactions as it cultivates mistrust towards digital transactions (Bandyopadhyay, 2012). It may be elaborated by observing the reluctance of the masses to switch to online banking or preferring cash-on-delivery payment rather than online payment for online shopping. Banks and financial bodies have to invest continually in e-safety awareness (Aloul, 2012), thus confirming the finding of this study.

4.3. Analysis of digital media usage enablers

As stated earlier, the secondary data is available for ten years only. However, the enablers are 4, implying that the number of observations is insufficient to estimate this model with all variables in one model (Hanke and Wichern, 2014). We overcome this issue by doing regression analysis using forward estimation (i.e., including one variable each in a model at a time). Thus, we have several regression models instead of one model, as shown in Table 3. This method does provide the impact of a single or few variables on the dependant variable (log of digital transactions in the economy). But, it fails to offer partial effects of all variables simultaneously. However, this is a limitation due to the lesser data availability on which the researchers have no control. LM test in Eview was used to check for autocorrelation. However, there was no autocorrelation issue in the estimated models. The results of the regression models are presented in Table 3.

The results of the first four models indicate a good fit with all individual variables being statistically significant with high R^2 values and statistically significant F -statistics with LM tests showing no autocorrelation issue. It means that overall, if we consider four hypotheses (H_9 to H_{12}) in isolation, then all of these are statistically confirmed. It implies a strong and statistically significant influence of digital media usage enablers (literacy rate, broadband subscriptions, internet usage, and mobile-phone subscriptions) on digital business transactions and sales.

4.3.1. Discussion on digital media usage enablers

The first model indicates that a 1% increase in literacy increases digital transactions by about 0.4%. Similarly, a 1% increase in mobile-

Table 3
Results of digital media usage enablers.

	Model 1	Model 2	Model 3	Model 4	Model 5
Variables	Coeff. (S. E)	Coeff. (S. E)	Coeff. (S. E)	Coeff. (S. E)	Coeff. (S. E)
Constant	−2.899 (4.425)	17.625*** (1.057)	17.618*** (0.416)	−70.549*** (11.551)	−43.718 (30.307)
Literacy rate (LI)	0.405*** (0.077)				−0.231 (0.282)
Broadband subscriptions (FB)		3.390** (1.295)			0.129 (0.589)
Internet Usage (IN)			0.214*** (0.031622)		0.156* (0.062)
Mobile-phone subscriptions (Ln(MB))				4.864*** (0.618)	4.025 (2.396)
Model fit information					
R ²	0.775	0.461	0.851	0.886	0.931
Adjusted-R ²	0.747	0.394	0.832	0.871	0.877
F – Statistics	27.613***	6.855***	45.685***	61.921***	16.981***
N	10	10	10	10	10
LM – Test	0.858	4.985	0.052	1.037	1.210
N*R ² ~ χ^2 (2)					

phone subscriptions increases digital transactions by about 5% (Model 2). Model 3 indicates that a one-unit increase in broadband subscriptions increase digital transactions by about 3%. Finally, Model-4 shows a one per cent increase in internet usage increases the digital transactions by about 0.2%. These findings are plausible.

Model 5 shows the combined model of the all-digital media usage enablers impacts on digital transactions. As expected, due to the smaller sample size, this model is not a good fit, despite having an excellent R² value, no autocorrelation issue with highly significant F-statistics value, only one variable (IN) is statistically significant. This model indicates that keeping other things constant (i.e., keeping LI, FB, and MB constant), if the internet access is increased by one unit, the digital transactions will increase by 0.15%. The impact of the other digital media usage enablers is statistically not significant, which may pertain to the low sample size.

5. Conclusion

This study aimed to explore how the digital economy of Pakistan can be strengthened using digital literacy. A theoretical framework was established using literature review and discussion with digital business owners. The theoretical framework was further operationalized by converting it into twelve workable hypotheses related to digital literacy competency and digital media usage enablers and their impact on digital business transactions. The study used primary and secondary data and employed a multiple linear regression model to test the hypothesis. The results confirmed that digital literacy and digital media usage enablers influence customer transactions and activity in the digital economy.

The results indicate that digital firms investment in customers various skills (such as functional skills, critical thinking, evaluation skills, collaborations skills, e-safety skills and problem-solving skills positively contribute) contributes to the firm business. These results are consistent with earlier studies like Bell & Eisingerich (2007) and Bell et al. (2017) that reported that educating customers about firm-specific skills may increase their loyalties (hence transactions) whereas, educating customers about market specific-expertise may lead to a decrease in their loyalty. However, our results indicate that firms' investment in customers' creative and communication skills does not enhance digital business sales that we believe are because of the unique nature of these skills that demand long term and consistent investment to develop.

Similarly, the findings that increase in literacy, mobile-phone subscriptions, broadband subscriptions, internet usage also increase business transactions. These findings are also consistent with These findings

are consistent with a more recent study of Hussain et al. (2021) that reported ICT penetration measured through both mobile phone and fixed phone subscriptions being positively linked with economic growth with earlier having more profound relation. It was concluded that Pakistani businesses penetrating the digital ecosystem could expand their market and increase customer transactions by investing in the digital literacy of their customers and potential target market. It was also inferred that empowering digital media usage could also convert to increased sales and digital economy transactions.

The study offers several theoretical and practical implications. For instance, the practical implication of the finding can be linked with digital economy development in Pakistan in the form of the establishment of various new apps and businesses in transportation (Careem, Bykea), food (Foodpanda) and social media (TikTok). Similarly, the study provides a foundation for studying the role of digital literacy role in enhancing firms business transactions.

It is essential to mention the limitation of having a smaller sample size for the digital media usage enablers. At the same time, it is equally vital to highlight that some of the concepts of digital literacy need to be updated (for instance, the functional skills will completely wipe out with time, as it refers to the basic operation of a digital device). Therefore, it is crucial to consider such improvements for future research.

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